Internet Of Things (IoT)

Perancangan Sistem IoT untuk Pengendalian Listrik Ruangan Berbasis Sensor Penghitung Orang dan Deteksi Pergerakan

Tujuan:

1. Merancang tampilan awal sistem IoT yang menggunakan sensor penghitung orang dan sensor deteksi gerak.
2. Menyusun cara kerja sistem otomatis untuk mengatur listrik di dalam ruangan.
3. Menggabungkan beberapa sensor ke dalam satu sistem IoT yang bekerja secara teratur.
4. Membuat simulasi jalannya sistem dalam kondisi ruangan sedang dipakai dan tidak dipakai.

Software Aplikasi

1. Arduino IDE versi 2.3.7

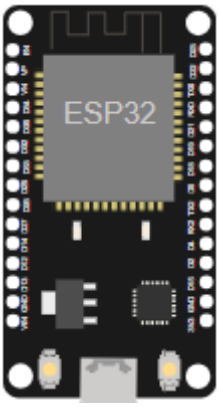
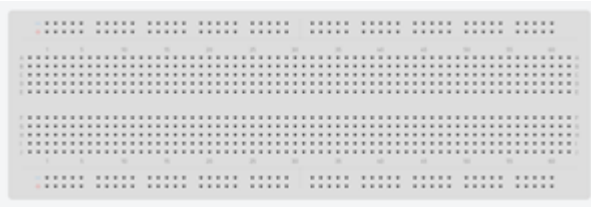
   software Arduino IDE (Integrated Development Environment) digunakan untuk menulis program (coding) dan mengunggah kode ke ESP32 board

2. Blynk IoT

   Platform Internet of Things (IoT) yang dapat diakses di situs https://blynk.io. Platform ini memungkinkan pengguna untuk menghubungkan dan mengontrol perangkat

List Komponen & Spesifikasi

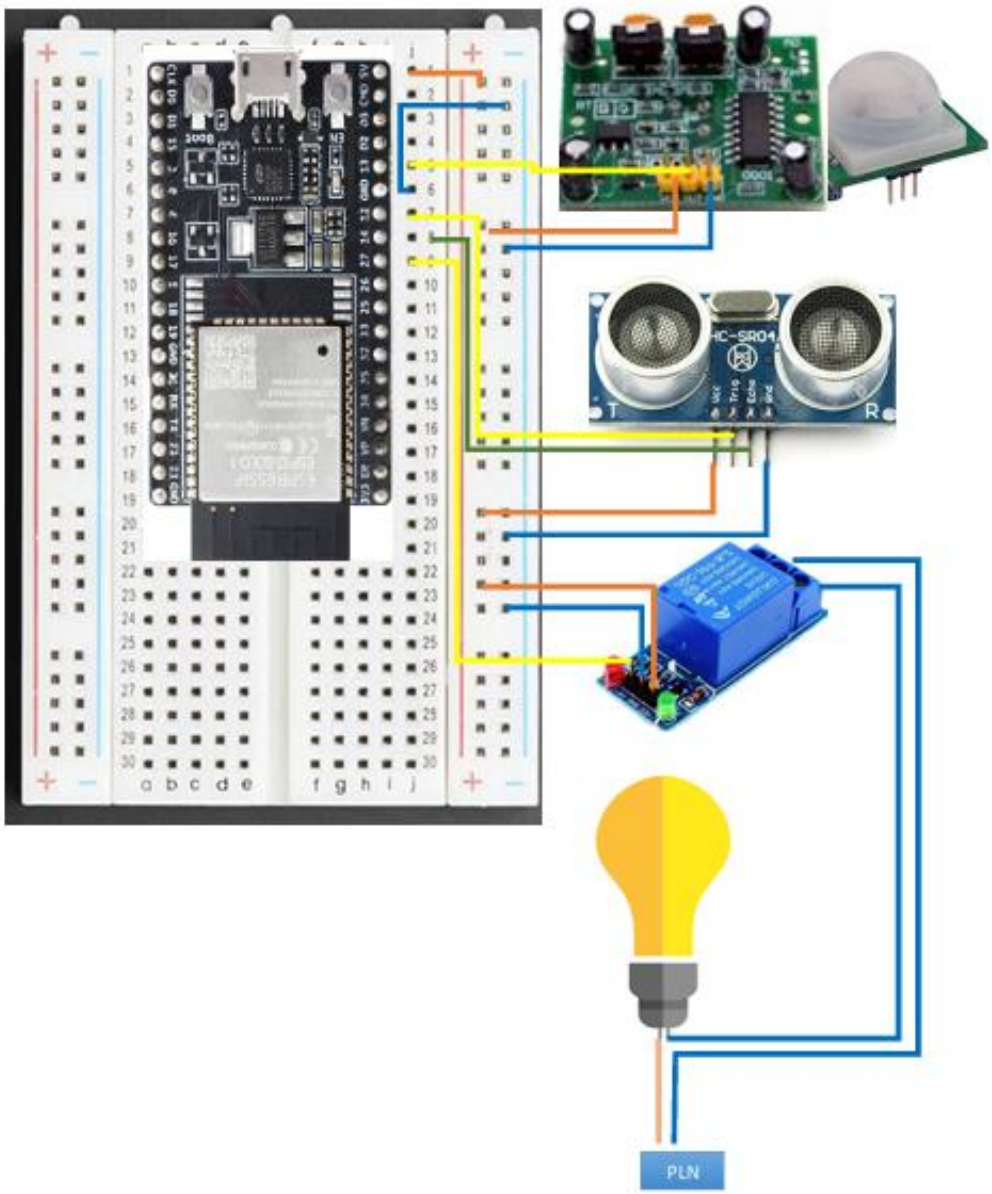
List komponen dan spesifikasi yang di gunakan adalah:

KOMPONEN	KETERANGAN DAN SPESIFIKASI
	1 buah ESP32 Development Board Spesifikasi: ● Operating voltage : 3.3V ● Input voltage : 7-9V (Vin) --> (Ke pin Vin, jangan ke Vcc) ● Digital IO Pin (DIO) : 25 ● Analog Input Pin (ADC) : 6 ● Analog Output Pin (DAC) : 2 ● UART : 3 ● SPI : 2 ● I2C : 3 ● Flash Memory 4 MB ● SRAM : 520 KB ● Clock Speed : 240 Mhz ● Wi Fi : IEEE 802.11 b/g/n/e/i ● Mode supported : AP, STA, AP+STA ● CP2102 USB ccontroller ● USB TYPE Mikro
	1 buah breadboard Spesifikasi: ● MB-102 MB102 Breadboard 830 Point Solderless PCB Bread project Board

	1 buah sensor Ultrasonic HC-SR04 Spesifikasi: ● Tegangan Kerja : 4,5 Volt s/d 5,5 Volt (Support Arduino) ● Jarak deteksi : 2 CM s/d 450 CM ● Efektif : 3 CM - 300 CM (Faktor lingkungan / halangan lain) ● Kepresisian : 0.3cm ● Area Buta : 2cm ● Arus Statik : < 2mA ● Frekuensi : 40 Khz ● Suhu operasi : -20 s/d -80 derajat ● Dimensi : (P x L x T) = 4,6cm x 2,2cm x 1,3cm Wiring / Pemasangan : ● VCC, trig (control side) ● Echo (receiving end) ● GND
	1 buah PIR Motion Sensor Spesifikasi: ● Panjang: 24.03mm/0.94in ● Lebar: 32.34mm/1.27in ● Jarak antar lubang: 28mm ● Diameter lubang: 2mm ● Tinggi dengan lensa: 24.66mm/0.97in ● Rentang sensitivitas: hingga 6 meter, 110 x 70 area deteksi ● Power supply: 3V-9V input voltage, idealnya 5v.

	1 buah relay Spesifikasi: ● Maximum load: AC 250V/10A, DC 30V/10A ● channel: 1 ● Working voltage: 5V, active LOW atau active HIGH ● Weight: 18 g ● Terdapat 2 indikator led yaitu indikator module dan indikator Relay aktif
	1 buah fitting lampu Spesifikasi: ● Fitting : E27
	1 buah Bohlam LED 4 W Spesifikasi: ● Merek : PHILIPS ● Watt : 4 Watt ● Volt : 220 Volt ● Cahaya : Putih ● Fitting : E27

Lampiran A – Diagram Blok Sistem IoT



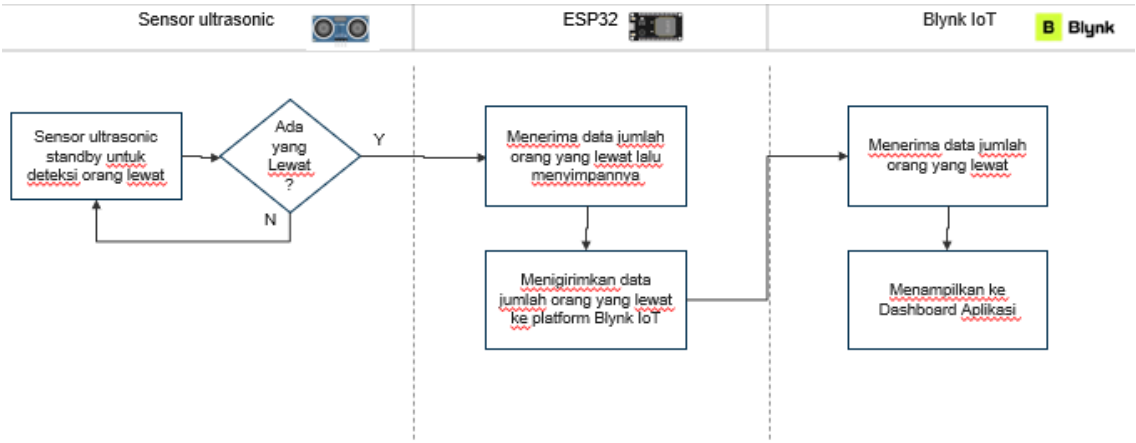
Gambar A.1 Diagram Blok Sistem IoT

Relay	PIR Motion	HC-SR042	ESP32	
VCC	VCC	VCC	VN	3V3
GND	GND	GND	GND	GND
	Out		D13	D15
		Echo	D12	D2
		Trig	D14	D4
In			D27	RX2
			D26	TX2
			D25	D05
			D33	D18
			D32	D19
			D35	D21
			D34	RX0
			VN	TX0
			VP	D22
			EN	D23

Gambar A.2 Wiring Diagram Sistem IoT

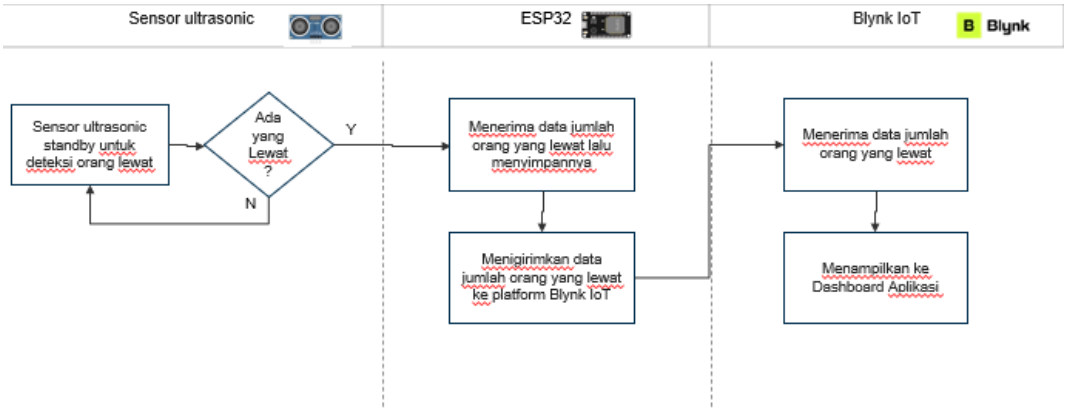
Lampiran B – Flowchart Sistem Pengendalian Listrik

1. Flowchart Sensor Penghitung Orang



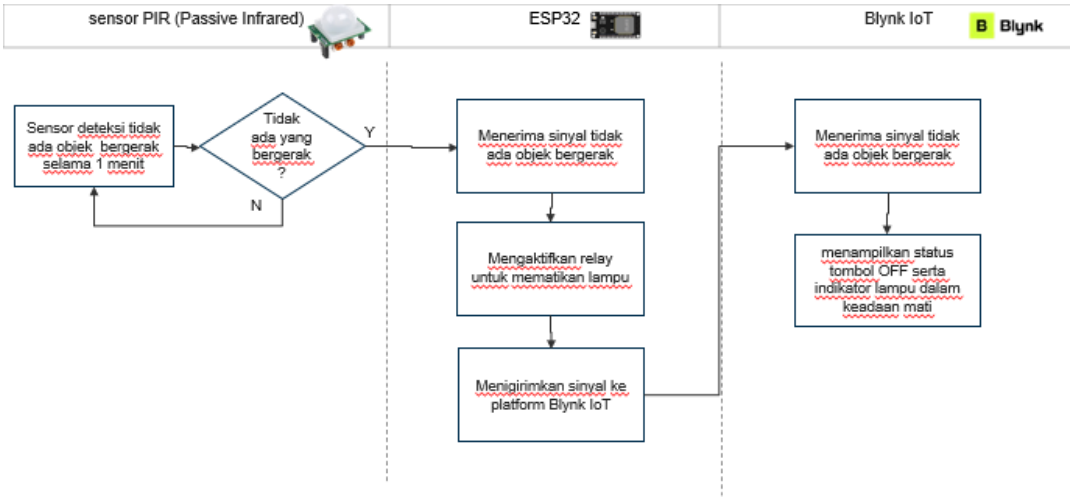
Gambar B.1 Flowchart Sensor Penghitung Orang

2. Flowchart Sistem Menyalakan Listrik Saat Ada Orang



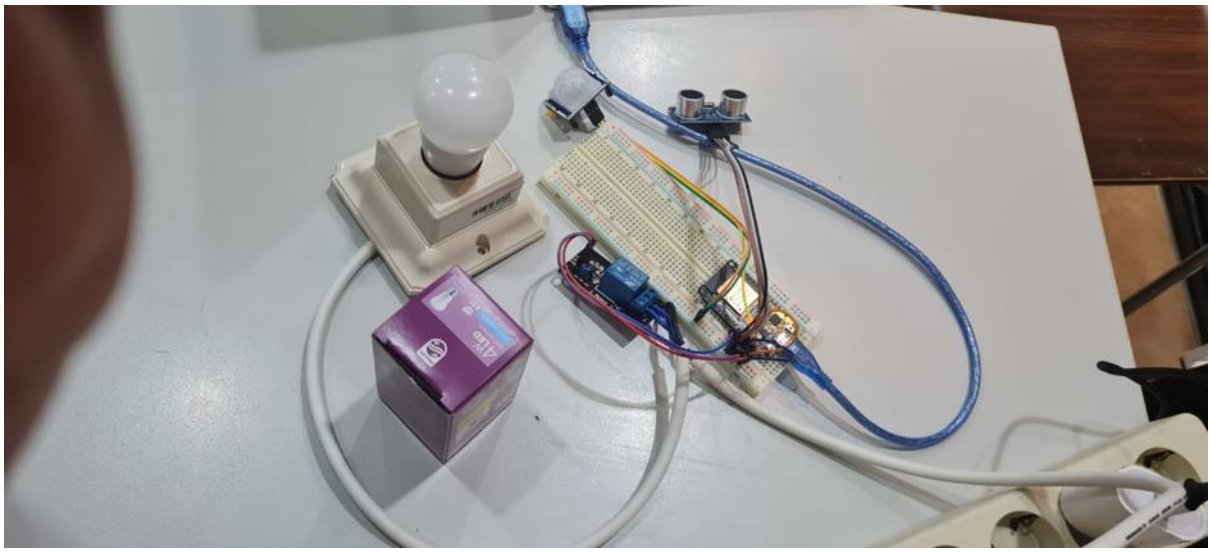
Gambar B.2 Flowchart Sistem Menyalakan Listrik Berdasarkan Deteksi Pergerakan

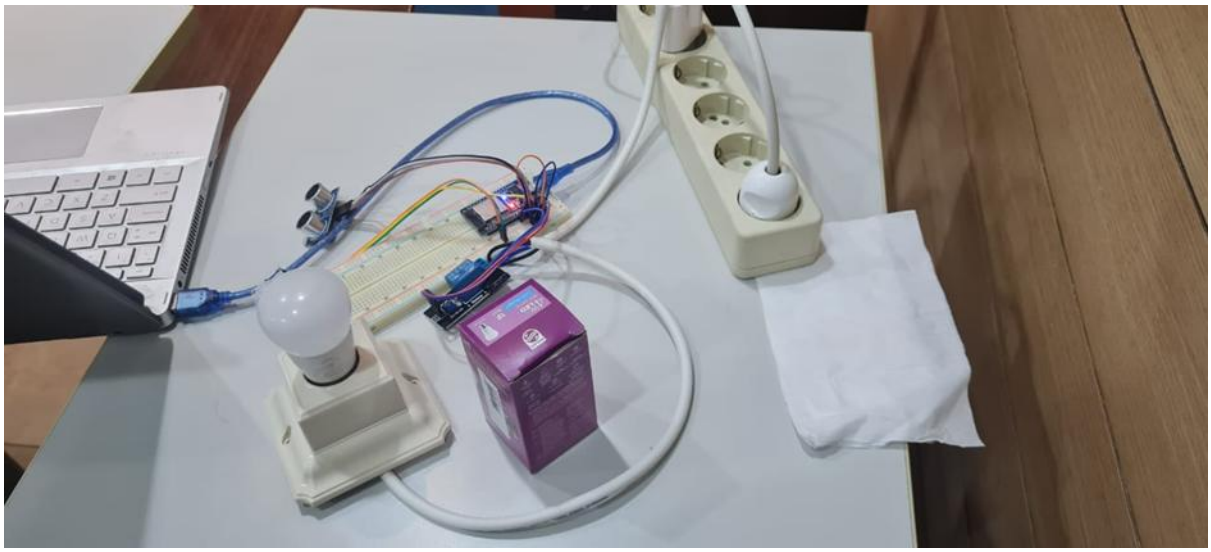
3. Flowchart Sistem Mematikan Listrik Saat Tidak Ada Orang

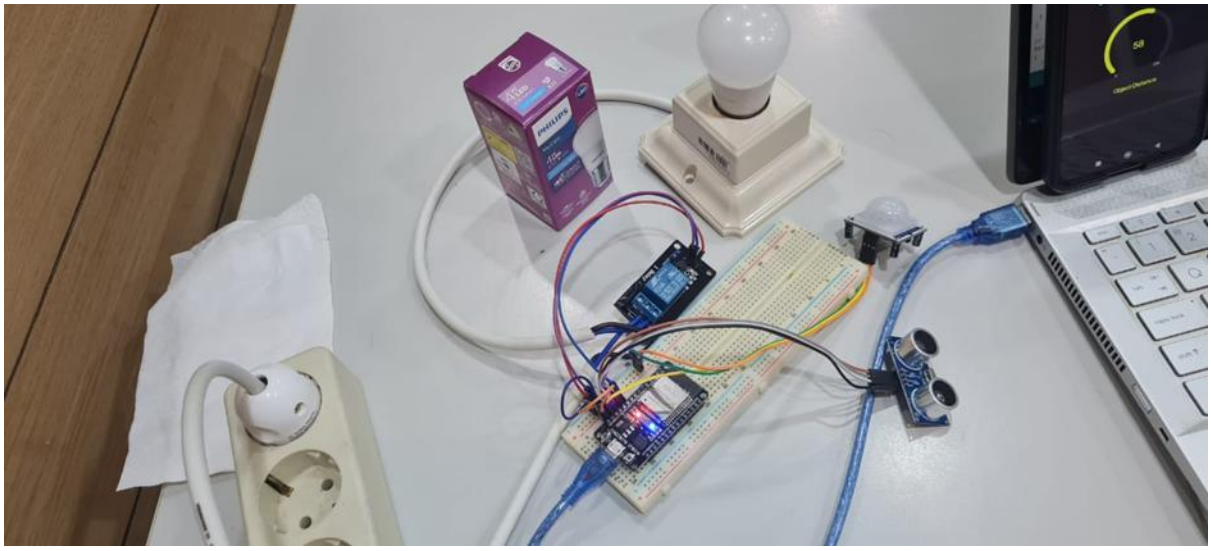


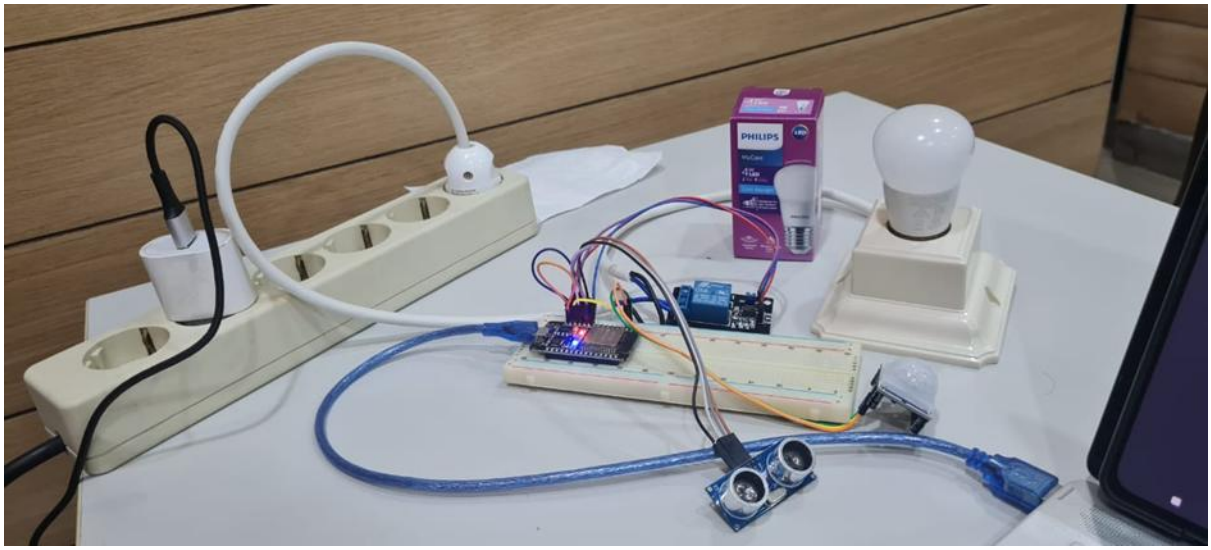
Gambar B.3 Flowchart Sistem Mematikan Listrik Saat Tidak Terdeteksi Pergerakan

Lampiran C – Dokumentasi Mock-up Sistem dari beberapa angel

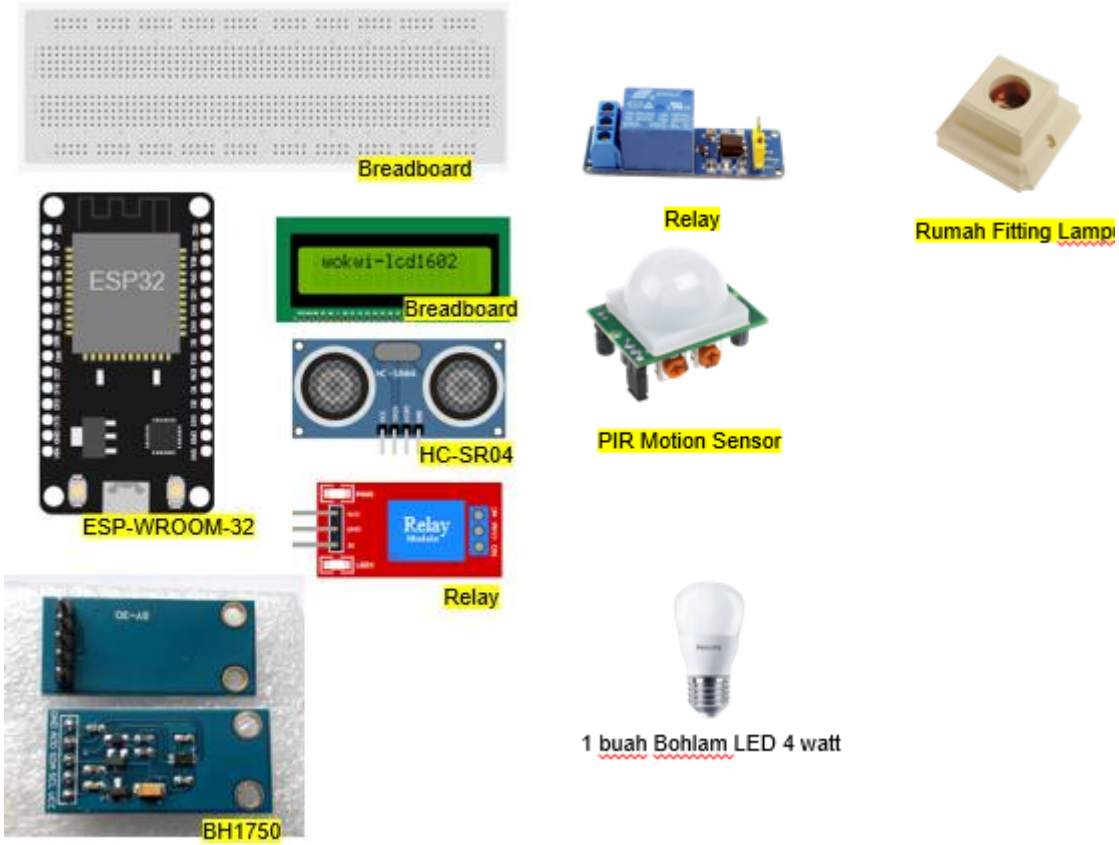








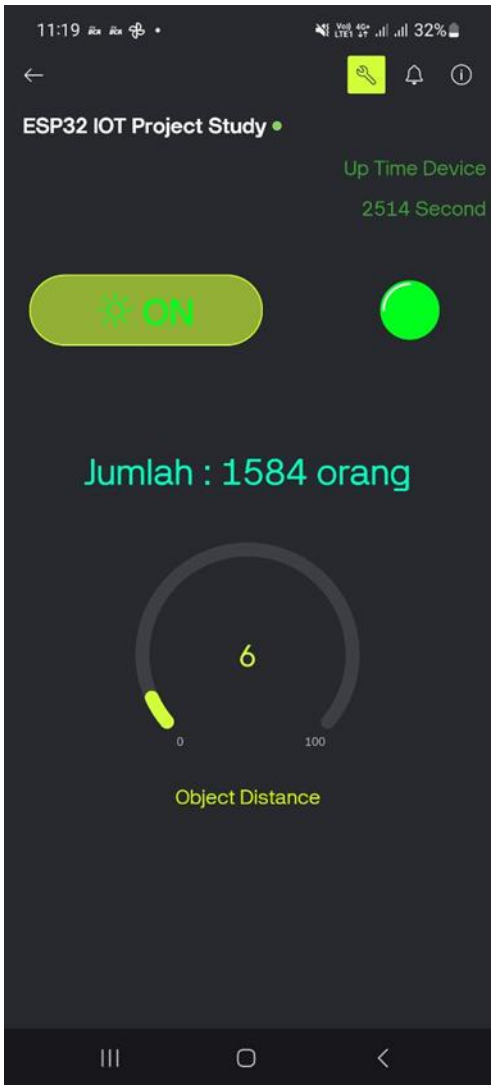
Gambar C.1 Mock-up Sistem IoT Tampak Keseluruhan



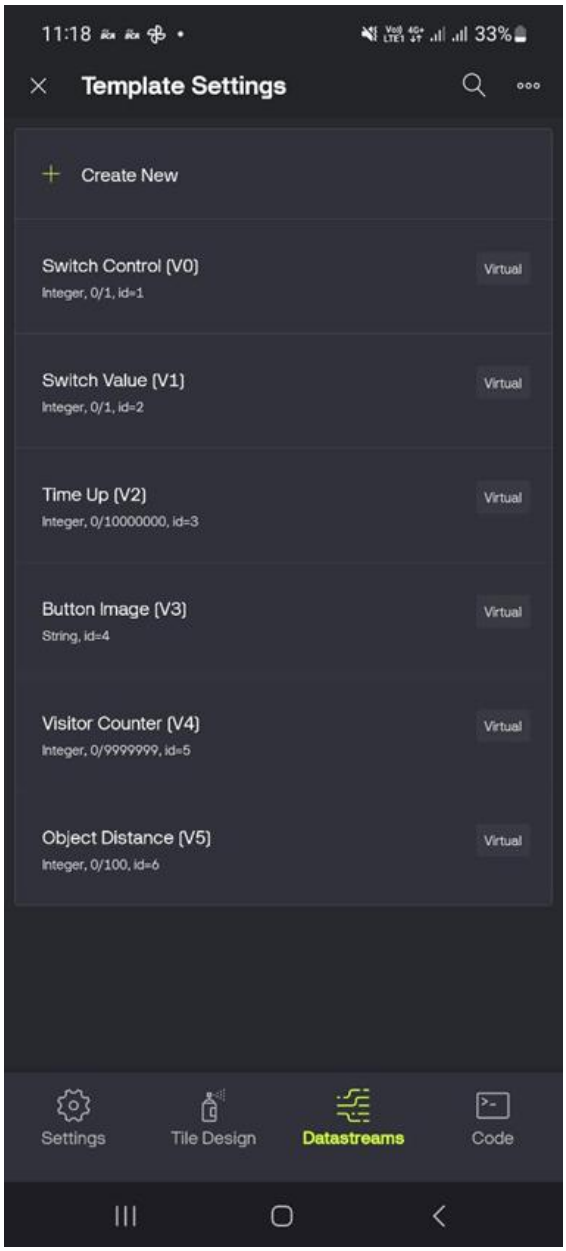
Gambar C.2 Detail Komponen Mock-up

Dokumentasi Mock-up Dashboard Aplikasi by BLINK

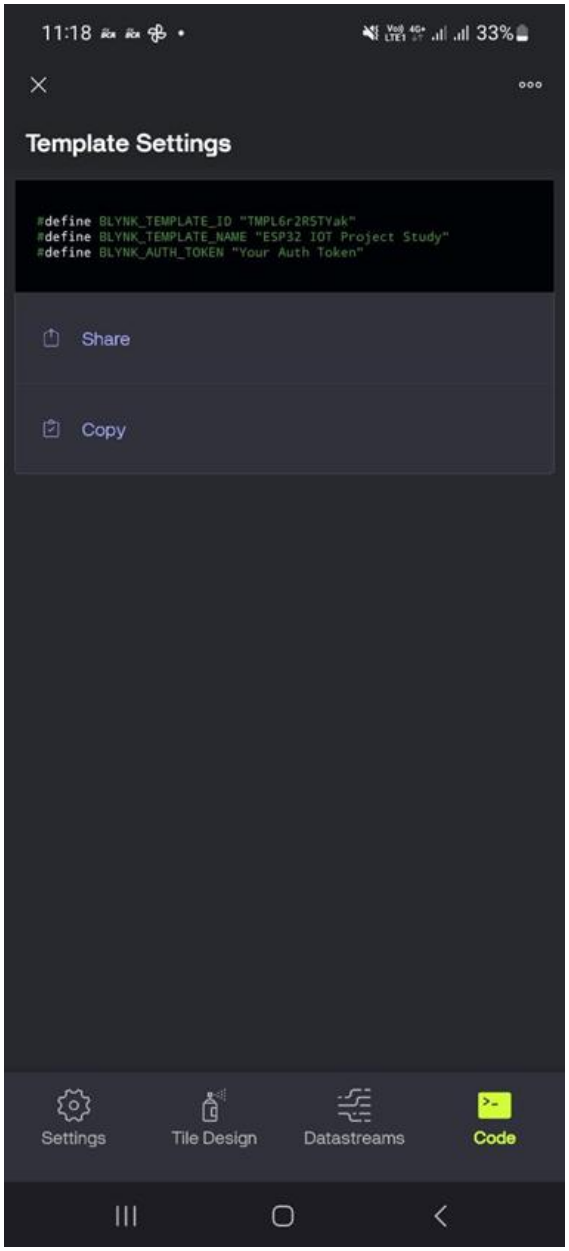
Gambar Dashboard dengan Lampu Off



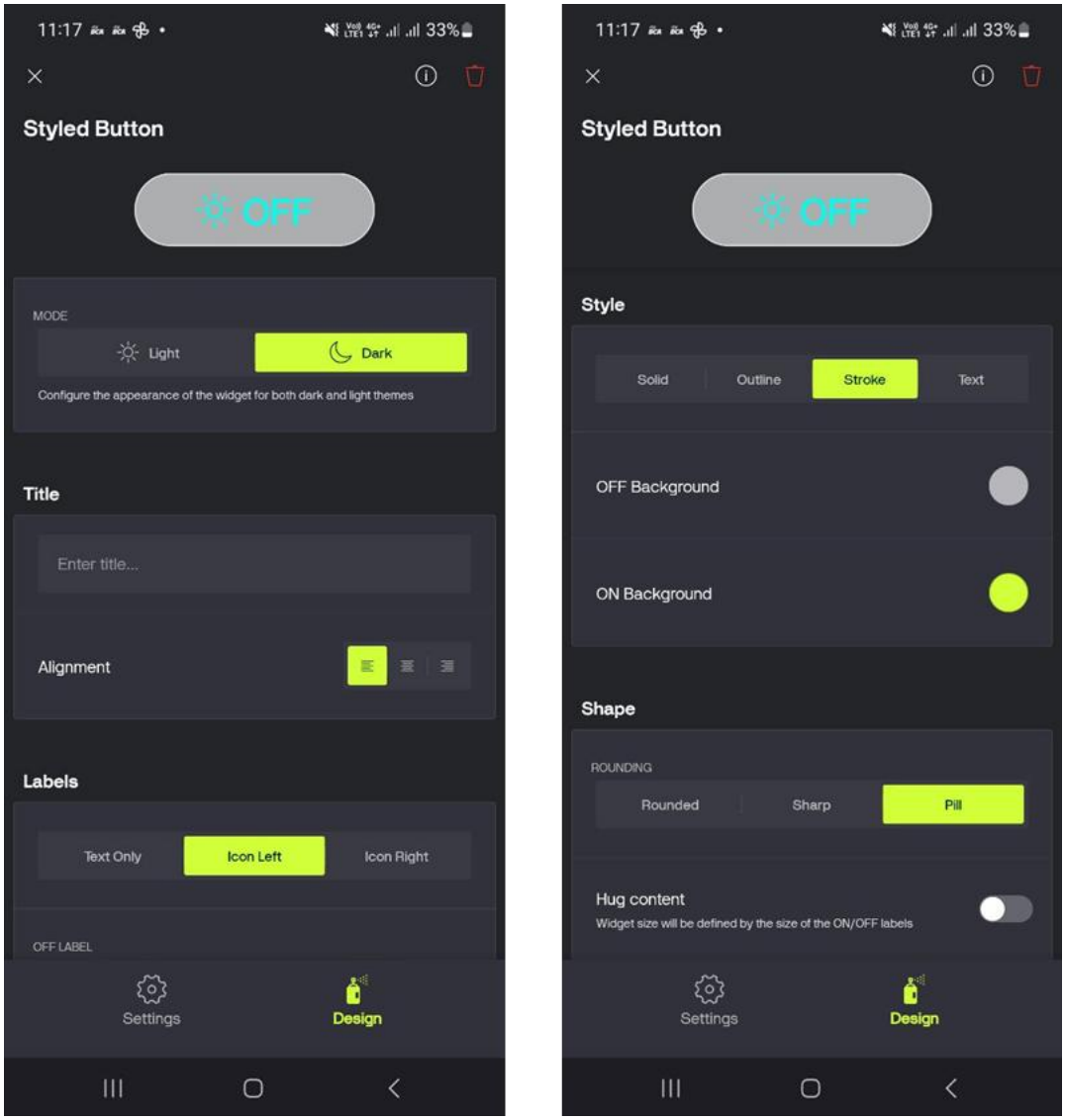
Gambar Dashboard dengan Lampu On



Gambar Dashboard Data Stream



Gambar Dashboard Code



Gambar Dashboard Design Style